

A. GENERAL NOTES

1. ALL NEW CONSTRUCTION SHALL COMPLY WITH THE CONTRACT DOCUMENTS AND THE INTERNATIONAL BUILDING CODE IBC / DUBAI BUILDING CODE.
2. REFERENCE TO CODES, RULES, REGULATIONS, STANDARDS, MANUFACTURER'S INSTRUCTIONS OR REQUIREMENTS OF REGULATORY AGENCIES ARE TO THE LATEST PRINTED EDITION OF EACH IN EFFECT AT THE DATE OF SUBMISSION OF BID UNLESS THE DOCUMENT DATE IS SHOWN.
3. THE STRUCTURAL DRAWINGS ILLUSTRATE THE NEW STRUCTURAL MEMBERS. REFER TO ARCHITECTURAL, MECHANICAL, AND ELECTRICAL DRAWINGS FOR NON-STRUCTURAL ITEMS WHICH REQUIRE SPECIAL PROVISIONS DURING THE CONSTRUCTION OF THE STRUCTURAL MEMBERS.
4. REFER TO ARCHITECTURAL DRAWINGS FOR FLOOR DEPRESSIONS, EDGE OF SLAB, OPENINGS, SLOPES, DRAINS, CURBS, PADS, EMBEDDED ITEMS, NON-BEARING PARTITIONS, ETC. REFER TO MECHANICAL AND ELECTRICAL DRAWINGS FOR SLEEVES, OPENINGS, AND HANGERS FOR PIPES, DUCTS AND EQUIPMENT.
5. ALL DIMENSIONS ARE IN MILLIMETRES U.N.O. ALL DIMENSIONS MUST BE VERIFIED ON SITE AND ANY DISCREPANCIES SHALL BE BROUGHT TO THE ATTENTION OF THE ENGINEER BEFORE PROCEEDING WITH THE PORTION OF WORK INVOLVED.
6. ALL ASTERISKED DIMENSIONS (INDICATED THUS: 123*) ARE TO BE CONFIRMED ON SITE AND REPORTED TO THE CONSULTANT FOR CONSIDERATION PRIOR TO COMMENCEMENT OF CONSTRUCTION.
7. ALL ELEMENTS OF THE STRUCTURE DESIGNED BY THE CONTRACTOR SHALL BE SUBMITTED TO THE CONSULTANT FOR APPROVAL PRIOR TO COMMENCEMENT OF CONSTRUCTION.
8. ALL MATERIALS SHALL BE APPROVED BY THE ENGINEER IN WRITING.
9. ALL TESTING SHALL BE AT THE CONTRACTOR'S EXPENSE AND SHALL BE CARRIED OUT AT AN INDEPENDENT TESTING LABORATORY APPROVED BY THE ENGINEER AND DUBAI MUNICIPALITY.
10. ALL DRAINAGE AND OTHER UNDER GROUND SERVICES SHALL BE LAID BEFORE CASTING OF THE GROUND BEARING SLAB & RAFTS
11. EXISTING CONDITIONS AS SHOWN ON THESE PLANS ARE FOR REFERENCE ONLY. CONTRACTOR IS REQUIRED TO FIELD VERIFY ALL EXISTING CONDITIONS PRIOR TO CONSTRUCTION. CONTRACTOR SHALL REPORT CONDITIONS THAT CONFLICT WITH THE CONTRACT DOCUMENTS TO THE OWNER'S REPRESENTATIVE. DO NOT DEVIATE FROM THE CONTRACT DOCUMENTS WITHOUT WRITTEN DIRECTION FROM THE OWNER'S REPRESENTATIVE.
12. THE CONTRACTOR SHALL VERIFY AND BE RESPONSIBLE FOR COORDINATING THE WORK OF ALL TRADES AND SHALL VERIFY ALL DIMENSIONS AND CONDITIONS WHICH IMPACT THE WORK. FIELD VERIFY SIZES, ELEVATIONS, HOLE LOCATIONS, ETC. PRIOR TO FABRICATION. THE LANGUAGE "BY OTHERS" USED IN THIS STRUCTURAL DRAWING SET INDICATES ELEMENTS OR PARTS OF WORK NOT WITHIN CONSULTANT SCOPE AND SHOWN OR REFERENCED FOR EASE OF COORDINATION ONLY. SUCH LANGUAGE SHALL NOT IMPLY THAT SUCH ELEMENTS OR PARTS OF WORK ARE EXCLUDED FROM THE CONTRACTOR'S SCOPE OF WORK.
13. THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE PROPER SETTING OUT OF THE WORKS, FOR CORRECTNESS OF LINE AND LEVEL AND FOR QUALITY CONTROL OF THE MATERIALS. THE APPROVAL OF THE ENGINEER SHALL NOT IN ANY WAY RELIEVE THE CONTRACTOR OF HIS RESPONSIBILITY AND ANY ERRORS SHALL BE RECTIFIED BY THE CONTRACTOR TO THE APPROVAL OF THE ENGINEER AT NO ADDITIONAL COST TO THE CONTRACT.
14. CONTRACTOR SHALL CAREFULLY REVIEW THE DRAWINGS TO IDENTIFY THE SCOPE OF WORK REQUIRED, VISIT THE SITE TO RELATE THE SCOPE OF WORK TO EXISTING CONDITIONS AND DETERMINE THE EXTENT TO WHICH THOSE CONDITIONS AND PHYSICAL SURROUNDINGS WILL IMPACT THE WORK.
15. THE CONTRACTOR SHALL RESOLVE ANY CONFLICTS ON THE DRAWINGS OR IN THE SPECIFICATIONS WITH THE OWNER'S REPRESENTATIVE BEFORE PROCEEDING WITH THE WORK.
16. ANY DEVIATION, MODIFICATION & SUBSTITUTION FROM THE APPROVED SET OF STRUCTURAL DRAWINGS SHALL BE SUBMITTED TO THE ENGINEER, PROJECT MANAGEMENT, AND OWNER'S REPRESENTATIVE FOR REVIEW/APPROVAL PRIOR TO ITS USE OR INCLUSION ON THE SHOP DRAWINGS & PRIOR TO PROCEEDING WITH THE WORK.
17. THE CONTRACTOR SHALL PROVIDE ALL NECESSARY CONCRETE FORMWORK SHORING-RE-SHORING, EXCAVATION SHORING, DEMOLITION SHORING, BRACES, CLIPS, HOIST BEAMS, ETC. REQUIRED TO SUPPORT ANY AND ALL LOADS THE BUILDING STRUCTURE AND COMPONENTS, EARTHWORK, OTHER STRUCTURES, AND UTILITIES ARE SUBJECTED TO DURING CONSTRUCTION. CONCRETE FORMWORK/CONCRETE RE-SHORING, DEMOLITION, AND EXCAVATION SHORING SYSTEMS MUST BE DESIGNED AND STAMPED STRUCTURAL ENGINEER LICENSED BY DUBAI MUNICIPALITY FOR THIS TYPE OF WORKS AND HIRED BY THE CONTRACTOR. CONSULTANT INSPECTIONS TO THESE WORKS SHOULD NOT RELIEF THE CONTRACTOR AND/OR HIS SPECIALIST SUBCONTRACTOR FROM THE RESPONSIBILITY.
18. THE CONTRACTOR SHALL PROVIDE MEANS, METHOD, TECHNIQUES, SEQUENCE AND PROCEDURE OF CONSTRUCTION AS REQUIRED. CONSULTANT INSPECTIONS TO THESE WORKS SHOULD NOT RELIEF THE CONTRACTOR AND/OR HIS SPECIALIST SUBCONTRACTOR FROM THE RESPONSIBILITY.
19. THE CONTRACTOR SHALL PROTECT ALL WORK, MATERIALS AND EQUIPMENT FROM DAMAGE AND SHALL PROVIDE PROPER STORAGE FACILITIES FOR MATERIALS AND EQUIPMENT DURING CONSTRUCTION.
20. ATTACHMENT OF NON-STRUCTURAL COMPONENTS SPECIFIED BY OTHERS TO STRUCTURAL ELEMENTS SHALL BE SPECIFIED BY THE NON-STRUCTURAL COMPONENT DESIGNER OR SPECIFIER OR INSTALLER. DESIGNER OF NON-STRUCTURAL ELEMENTS SHALL AT A MINIMUM SPECIFY THE CONNECTION TO THE STRUCTURE INCLUDING BUT NOT LIMITED TO: ANY TYPE OF CONNECTING HARDWARE, WIRE, HANGERS, FASTENERS, CLIPS, UNISTRUT MEMBERS. ATTACHMENT AND BRACING OF NON STRUCTURAL COMPONENTS SHALL MEET THE APPLICABLE BUILDING CODES. NON STRUCTURAL ELEMENTS SHALL INCLUDE, BUT NOT LIMITED TO: MEP AND HVAC EQUIPMENT & THEIR SUPPORTING PADS, INDUSTRIAL KITCHEN EQUIPMENT, PLATFORMS, FRAMES, ETC.; DUCTWORK, PIPES, CONDUITS, ARTWORK, GRILLES, GRATING, METAL SCREENS, ELEVATOR RAILS, STONE FINISH TILES, STONE CAPS, BRICK VENEER.
21. GENERAL CONTRACTORS AND SUBCONTRACTORS SHALL REVIEW AND PROVIDE APPROVAL STAMP FOR ALL STRUCTURAL SHOP DRAWINGS AND SUBMITTALS PRIOR TO SUBMITTING TO CONSULTANT.
22. ALLOW 10 BUSINESS DAYS FOR PROCESSING SHOP DRAWINGS AND SUBMITTALS AFTER RECEIPT. ALLOW 5 BUSINESS DAYS FOR RESPONDING TO REQUESTS FOR INFORMATION (RFIs). PROVIDE 3 BUSINESS DAYS NOTICE FOR STRUCTURAL INSPECTION.
23. STRUCTURAL DRAWINGS SHALL BE READ IN CONJUNCTION WITH SPECIFICATIONS AND ALL OTHER CONTRACT DOCUMENT.
24. THE CONTRACTOR SHALL PROVIDE ANTI-TERMITE TREATMENT AROUND THE BUILDING IN ACCORDANCE WITH LOCAL REQUIREMENTS AS OUTLINED IN SPECIFICATION.

B. CONCRETE

1. CONCRETE IS REINFORCED AND CAST-IN-PLACE UNLESS OTHERWISE NOTED. WHERE REINFORCING IS NOT SPECIFICALLY SHOWN OR WHERE DETAILS ARE NOT GIVEN, PROVIDE REINFORCING SIMILAR TO THAT SHOWN FOR SIMILAR CONDITIONS. SUBJECT TO REVIEW BY THE ENGINEER.
2. ALL PHASES OF WORK PERTAINING TO THE CONCRETE CONSTRUCTION SHALL CONFORM TO THE LATEST EDITION OF ACI 318 "BUILDING CODE REQUIREMENTS FOR STRUCTURAL CONCRETE" AND THE LATEST EDITION OF ACI 117 "SPECIFICATIONS FOR TOLERANCES FOR CONCRETE CONSTRUCTION AND MATERIALS"
3. CONCRETE MIXES SHALL MEET FOLLOWING SPECIFICATIONS:

CONCRETE MIX SPECIFICATIONS							
LOCATION	WEIGHT	COMPRESSIVE STRENGTH (CYLINDER/ CUBE) (N/mm ²)	MAX. WIC RATIO	SLUMP (mm)	LARGEST Dagg (mm)	MAXIMUM SHRINKAGE @ 28 DAYS	MINIMUM COMBINATION CONTENT (kg/m ³)
SUBSTRUCTURE							
FOUNDATION	NORMAL	C36/45	0.35	200 +/- 40	20	0.045%	380
BASEMENT WALL	NORMAL	C36/45	0.35	200 +/- 40	20	0.045%	380
SLAB ON GRADE	NORMAL	C36/45	0.35	200 +/- 40	20	0.045%	380
SCREED	NORMAL	C20/10	0.35	200 +/- 40	10	0.045%	380
SUPERSTRUCTURE							
CONCRETE COLUMN	NORMAL	C36/45	0.35	200 +/- 40	20	0.045%	380
CONCRETE SHEAR WALL	NORMAL	C36/45	0.35	200 +/- 40	20	0.045%	380
ELEVATED RC SLAB AND BEAMS	NORMAL	C36/45	0.35	200 +/- 40	20	0.045%	380
ALL LOCATIONS U.N.O.	NORMAL	C36/45	0.35	200 +/- 40	20	0.045%	380

4. NO MORE THAN ONE GRADE OF CONCRETE SHALL BE ON THE JOB SITE AT ANY ONE TIME.
5. ALL STRUCTURAL CONCRETE MIXES SHALL BE DESIGNED BY AN APPROVED LABORATORY AND SHALL BE REVIEWED AND APPROVED BY THE ENGINEER OF RECORDED.
6. CONCRETE MIX PROPORTIONING SHALL MEET STATISTICAL STRENGTH REQUIREMENTS OF ACI 301 AND ACI 214R. MIX DESIGNS SHOWING COMPLIANCE WITH STRENGTH REQUIREMENTS TO BE SUBMITTED TO THE ENGINEER FOR REVIEW.
7. CONCRETE STRENGTH TEST REPORTS SHALL BE IN COMPLIANCE WITH ACI 318 AND SHALL BE SUBMITTED TO THE ENGINEER.
8. CONCRETE MATERIALS AND MIXTURES:

A. MIXES SHALL BE PREPARED WITH TYPE I/IV PORTLAND CEMENT CONFORMING TO ASTM C150.

B. FLY ASH CONFORMING WITH ASTM C618 MAY REPLACE PORTLAND CEMENT CONTENT BY WEIGHT, UNLESS OTHERWISE NOTED ON THE CONCRETE MIX SPECIFICATIONS. REPLACEMENT MAY BE UP TO 15% FOR ELEVATED SLAB OR 25% FOR OTHER CASES. WHERE CONCRETE IS VISUALLY EXPOSED VERIFY WITH THE PROJECT ARCHITECT THE USE OF FLY ASH.

C. NORMAL WEIGHT CONCRETE AGGREGATES SHALL CONFORM TO ASTM C33. LIGHT WEIGHT CONCRETE AGGREGATES SHALL CONFORM TO ASTM C330, UNLESS OTHERWISE NOTED ON THE CONCRETE MIX SPECIFICATIONS. COARSE AGGREGATE GRADATION SHALL INCLUDE LARGEST AGGREGATE OF 25MM, EXCEPT THAT A LARGEST SIZE OF 10MM IS ALLOWED FOR FOUNDATION, COLUMNS, AND WALLS.

D. WATER USED IN MIXING CONCRETE SHALL CONFORM WITH ASTM C1602.

E. ADMIXTURES SHALL COMPLY WITH ASTM STANDARDS NOTED ON LATEST EDITION OF ACI 318.

F. ADDITIVES SPECIFIED BY OTHER CONSULTANT, SUCH AS, BUT NOT LIMITED TO, WATER PROOFING ADDITIVES, PIGMENTS, ETC, SHALL NOT IMPAIR THE EXPECTED STRUCTURAL PERFORMANCE OF THE CONCRETE. SPECIFICATIONS SHALL BE SUBMITTED TO CONSULTANT FOR REVIEW AND APPROVAL.
9. THOROUGHLY CLEAN AND ROUGHEN ALL HARDENED CONCRETE SURFACES TO RECEIVE NEW CONCRETE. INTERFACE SHALL BE ROUGHENED TO A FULL AMPLITUDE OF 6 MM WITH EXPOSED AGGREGATE UNLESS NOTED OTHERWISE.
10. DEFECTIVE CONCRETE (VOIDS, ROCK POCKETS, HONEYCOMBS, CRACKING, ETC.) SHALL BE REMOVED AND REPLACED AS DIRECTED BY THE OWNER'S REPRESENTATIVE.
11. KEY AND DOWEL POUR JOINTS AS SHOWN ON THE PLANS. ANY DEVIATION FROM POUR JOINTS SHOWN ON THE PLANS MUST BE APPROVED BY THE ENGINEER.
12. WHERE ELEMENTS SUCH AS, BUT NOT LIMITED TO, CONDUITS, PIPES, AND DUCTWORK, ARE TO BE PLACED WITHIN OR THRU CONCRETE MEMBERS, DIRECT CONTACT OF SUCH MEMBERS WITH CONCRETE SHALL BE PREVENTED AS REQUIRED BY DESIGNER/SPECIFIER/SUPPLIER/INSTALLER OF SUCH ELEMENTS, AND AS NEEDED TO COMPLY WITH PLUMBING CODE AND/OR TO AVOID DAMAGE OF SUCH ELEMENTS. TYPICAL MEANS TO AVOID DIRECT CONTACT WITH CONCRETE SUCH AS SLEEVES, LAYERS OF COMPRESSIBLE MATERIALS, AIR GAPS, ETC, SHALL BE SPECIFIED BY, OR OBTAINED BY GC FROM, DESIGNER/SPECIFIER/SUPPLIER/INSTALLER OF SUCH ELEMENTS. REBAR DISPLACED BY SUCH ELEMENTS SHALL BE ARRANGED AS CLOSE AS PRACTICALLY POSSIBLE AROUND THE EMBEDDED ELEMENTS AND SHALL NOT BE INTERRUPTED.
13. TO MINIMIZE CONCRETE SHRINKAGE CRACKING IN CONCRETE SLABS, THE MAXIMUM SIZE OF CONCRETE POURS FOR SLABS ON GRADE, FORMED SLABS, AND SLABS ON METAL DECK, IS 70 M IN ANY DIRECTION. THE RATIO OF THE PLAN LENGTH (LONGER DIRECTION) TO WIDTH (SHORTER DIRECTION) DIMENSIONS SHALL NOT EXCEED 2 TO 1.
14. FORMWORK REMOVAL:

A. FORMWORK SHALL BE REMOVED IN A MANNER THAT MAINTAINS THE STRENGTH AND STABILITY OF THE STRUCTURE AT ALL TIMES.

B. NO SHIPPING OF FORMWORK IS ALLOWED BEFORE CONCRETE HAS CURED SUFFICIENTLY TO PREVENT SPALLING, CHIPPING OR OTHER DAMAGE DUE TO FORMWORK REMOVAL.

C. NON PT SLABS: FORMWORK SHALL NOT BE REMOVED BEFORE ALL CONDITIONS BELOW ARE MET:

C.A. THE SPECIFIED COMPRESSIVE STRENGTH IS REACHED

C.B. FORM WORK IS LEFT IN PLACE AT LEAST 28 DAYS AFTER CONCRETE POUR, EXCEPT THAT FORM WORK CAN BE REMOVED 14 DAYS AFTER CONCRETE POUR IF FOLLOWED BY IMMEDIATE RE-SHORING OF THE SLAB ON THE SAME DAY. RE-SHORING SHALL BE MAINTAINED IN PLACE A MIN OF 28 DAYS AFTER CONCRETE POUR.

D. CONTRACTOR SHOULD SUBMIT A RE-SHORING CALCULATIONS TO ENSURE DURING CONSTRUCTION AND POURING OF CONCRETE THAT THE SUSPENDED SLABS ARE NOT LOADED WITH ANY EXTRA LOADS MORE THAN THE DESIGN LOADS SPECIFIED ON THESE DRAWINGS
16. CONCRETE PLACEMENT METHOD. GENERAL CONTRACTOR SHALL SELECT CONCRETE PLACEMENT METHOD, INCLUDING, BUT NOT LIMITED TO, POURED IN PLACE, SHOTCRETE, ETC., ADEQUATE TO COMPLY WITH ALL CONCRETE SPECIFICATIONS PER CONSTRUCTION DOCUMENTS, SUCH AS, BUT NOT LIMITED TO, CONCRETE MIXES PARAMETERS, ARCHITECTURAL FINISH, EXPOSED CONCRETE, ETC.
17. TO MINIMIZE SHRINKAGE AND CRACK IN BASEMENT, STAGGERED PANELS SHOULD BE POURED WITH A MAXIMUM WALL PANEL LENGTH OF 8 AND 10 METERS, FOR WALLS INSTALLED DIRECTLY AGAINST SHORING AND WALLS FORMED FROM BOTH FACES, RESPECTIVELY.
18. CORING AND DRILLING THRU, AND SAW CUTTING OF, CONCRETE ELEMENTS SHALL NOT BE PERMITTED WITHOUT CONSULTANT APPROVAL. PROVIDE REBAR SCANNING FOR CONSULTANT REVIEW.
19. CONCRETE FINISHES SHALL BE DESCRIBED IN THE SPECIFICATION AND ON THE ARCHITECT'S DRAWINGS AND SHALL GENERALLY BE SOUND AND FREE FROM MAJOR BLEMISH.
20. FOR HOT WEATHER CONCRETING REFER TO THE SPECIFICATION. THE INITIAL TEMPERATURE OF FRESH CONCRETE MUST NOT EXCEED 30°C. CONCRETING SHALL NOT TAKE PLACE IN AMBIENT SHADE TEMPERATURE EXCEEDING 43°C.
21. MASS CONCRETE BACK FILL IS TO BE USED AS INSTRUCTED BY THE ENGINEER.

22. LIGHT WEIGHT CONCRETE FILL SHALL BE A DENSITY OF 10.0 kN/m³ UNO.
23. FILL MATERIAL EXPANDED POLYSTYRENE GEOFOAM, EPS SHALL BE USED FOR LANDSCAPES BUILD UP FILL MATERIAL. EXPANDED POLYSTYRENE GEOFOAM SHALL BE IN COMPLIANCE WITH ASTM D6817&C-578 WITH MAXIMUM DENSITY OF 50kg/m OF 40 kPa AT 1% DEFORMATION-ANTI TERMITE. EPS SHALL BE TOPPED BY DENSE SCORED 100MM OR 150MM WITH ANTI CRACK MESH A252 (IF NEEDED) CONFINED BY PRIMARY STRUCTURES OR UP STANDS TO BE ADDED AS REQUIRED TO THE EXACT EXTENT AS PER ARCHITECTURAL & LANDSCAPE DRAWINGS.
24. CONSTRUCTION JOINTS LOCATIONS, PATTERN AND DETAILS SHALL BE PREPARED BY THE CONTRACTOR AND SUBMITTED ALONG WITH STRUCTURAL CALCULATIONS FOR ENGINEER'S APPROVAL. CONTRACTOR SHALL SUBMIT PLANS SHOWING THE POUR STRIPS LOCATION AND DETAILS. POUR STRIPS SHALL BE LIFT FOR AT LEAST 2 WEEKS BEFORE FILLING WITH CONCRETE.
25. IN THE ABSENCE OF ANY SPECIFIC CAMBER, THE SOFFIT OF ALL BEAMS/SLABS MORE THAN 700mm IN SPAN SHALL BE LAID TO CAMBER=L/750, WHERE L IS THE MAX. SPAN LENGTH OF TWO WAY SLAB OR SPAN LENGTH OF ONE WAY SLAB OR BEAM MEASURED FROM THE FACE OF THE SUPPORT.
26. IN THE ABSENCE OF ANY SPECIFIC CAMBER, THE SOFFIT OF ALL CANTILEVER BEAMS/SLABS MORE THAN 2000mm IN LENGTH SHALL BE LAID TO CAMBER=L/400, WHERE Lc IS THE SPAN LENGTH OF THE CANTILEVER BEAM/SLAB MEASURED FROM THE FACE OF THE SUPPORT.
27. IN THE ABSENCE OF ANY SPECIFIC CAMBER THE SOFFIT OF ALL STEEL BEAMS MORE THAN 1000MM IN SPAN SHALL BE LAID TO CAMBER 1/500 OF THEIR LENGTH AT THEIR CENTER TO SUPPORT.

C. CONCRETE PROTECTION

1. ALL BELOW GROUND CONCRETE STRUCTURES (INCLUDING BASEMENT WALLS INSTALLED DIRECTLY AGAINST SHORING OR OTHER BLIND SIDE INSTALLATIONS) AND OTHER CONCRETE STRUCTURES IN CONTACT WITH THE GROUND SHALL BE WATERPROOFED / PROTECTED WITH A CONTINUOUS WATERPROOFING MEMBRANE SYSTEM IN ACCORDANCE WITH THE PROJECT SPECIFICATIONS AND MANUFACTURER'S RECOMMENDATIONS.
2. ALL ASSOCIATED WATERPROOFING FITTINGS SHALL BE IN ACCORDANCE WITH THE RELEVANT SECTIONS OF THE PROJECT SPECIFICATIONS.
3. THE WATERPROOFING MEMBRANE SYSTEMS AND ASSOCIATED WATERPROOFING FITTINGS SHALL HEREIN BE TERMED THE ENTIRE WATERPROOFING SYSTEMS.
4. ALL WATERPROOFING MEMBRANE SYSTEMS DETAILS AND ASSOCIATED WATERPROOFING FITTINGS DETAILS SHALL BE IN ACCORDANCE WITH THE PROJECT SPECIFICATIONS AS NOTED ABOVE, THE GENERIC TYPICAL DETAIL DRAWINGS, THE DESIGN/SHOP DRAWING REQUIREMENTS AS STIPULATED BY THE WATERPROOFING SYSTEMS DESIGNER/CONTRACTOR AND THE ASSOCIATED MANUFACTURER'S DETAILS.
5. IN THE EVENT THAT THE WATERPROOFING SYSTEMS DESIGNER/CONTRACTOR PROPOSES AN ALTERNATIVE WATERPROOFING MEMBRANE SYSTEM AND/OR ASSOCIATED WATERPROOFING FITTINGS TO THE REQUIREMENTS OF THE PROJECT SPECIFICATIONS, THE SAME SHALL BE SUBMITTED TO THE ENGINEER FOR REVIEW AND APPROVAL PRIOR TO INCORPORATION INTO THE WORKS.
6. THE WATERPROOFING SYSTEMS DESIGNER/CONTRACTOR IS RESPONSIBLE FOR THE DESIGN AND IN-SERVICE PERFORMANCE OF ALL WATERPROOFING MEMBRANE SYSTEMS AND ASSOCIATED WATERPROOFING FITTINGS TO ACHIEVE THE DESIGN/PERFORMANCE PROVISIONS OF THE PROJECT SPECIFICATIONS.
7. WHERE DISCREPANCIES EXIST BETWEEN THE GENERIC TYPICAL DETAILS SETFORTH BY THE STRUCTURAL DRAWINGS AND THE DESIGN/SHOP DRAWING REQUIREMENTS AS STIPULATED BY THE WATERPROOFING SYSTEMS DESIGNER/CONTRACTOR, THE WATERPROOFING SYSTEMS DESIGNER/CONTRACTOR SHALL NOTIFY THE ENGINEER OF THE SAME AND OBTAIN APPROVAL OF THE PREFERRED DETAIL PRIOR TO INCORPORATING IN THE WORKS.
8. UPON APPROVAL RECEIPT FOR ANY SUCH OCCURRENCES, AS REQUIRED, THE WATERPROOFING SYSTEMS DESIGNER/CONTRACTOR SHALL ADVISE AND COORDINATE WITH ALL OTHER TRADES TO ENSURE THAT ALL ASPECTS ASSOCIATED WITH THE APPROVED DETAIL IS ACCOMMODATED IN THE WORKS.
9. THE WATERPROOFING SYSTEMS DESIGNER/CONTRACTOR SHALL DESIGN AND INSTALL ALL ASSOCIATED WATERPROOFING FITTINGS INCLUDING BUT NOT LIMITED TO WATERPROOFING DETAILS AT ALL PENETRATIONS (INCLUDING BUT NOT LIMITED TO ELEMENTS OF TEMPORARY WORKS, DEWATERING POINTS, STRUCTURAL CONNECTIONS, UTILITY PENETRATIONS), CONSTRUCTION JOINTS, WATER STOPS AND LAPS TO ENSURE WATERPROOFING PERFORMANCE FOR THE ENTIRE WATERPROOFING SYSTEMS ARE IN ACCORDANCE WITH THE REQUIREMENTS OF THE PROJECT SPECIFICATIONS.
10. THE WATERPROOFING SYSTEMS DESIGNER/CONTRACTOR SHALL ENSURE COMPATIBILITY OF ALL MATERIALS EMPLOYED FOR THE WATERPROOFING MEMBRANE SYSTEMS AND ASSOCIATED WATERPROOFING FITTINGS EMPLOYED IN THE ENTIRE WATERPROOFING SYSTEMS.
11. THE WATERPROOFING SYSTEMS DESIGNER/CONTRACTOR SHALL BE RESPONSIBLE FOR THE DESIGN AND CONSTRUCTION OF THE WORKS, AS APPROVED BY THE ENGINEER, ASSOCIATED WITH WATERPROOFING THE STRUCTURE AND SHALL FURNISH ALL LABOR, MATERIALS, TOOLS AND EQUIPMENT AND PERFORM ALL OPERATIONS NECESSARY FOR WATERPROOFING TO ACHIEVE THE SPECIFIED PERFORMANCE CRITERIA AS SETFORTH UNDER THE PROJECT SPECIFICATIONS.
12. THE WATERPROOFING SYSTEMS DESIGNER/CONTRACTOR SHALL ENSURE THAT THE ENTIRE WATERPROOFING SYSTEMS SHALL BE APPLIED IN ACCORDANCE WITH THE MANUFACTURER'S WRITTEN RECOMMENDATIONS BY AN INSTALLER CERTIFIED BY THE WATERPROOFING MANUFACTURER(S).
13. THE WATERPROOFING SYSTEMS DESIGNER/CONTRACTOR AND MANUFACTURER(S) SHALL CERTIFY AND WARRANT THE ENTIRE WATERPROOFING SYSTEMS WORKS IN ACCORDANCE WITH THE REQUIREMENTS OF THE PROJECT SPECIFICATIONS.

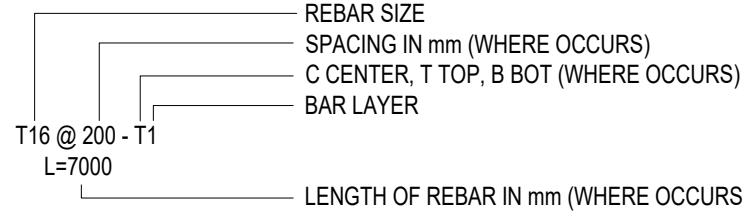
D. REINFORCEMENT

1. ALL REINFORCING BARS SHALL CONFORM TO BS4449 WITH MINIMUM YIELD STRENGTH OF 460MPa
2. REINFORCING STEEL SHALL HAVE THE FOLLOWING CONCRETE COVER. SEE ACI FOR TOLERANCES:

SPECIFIED CONCRETE COVER			
CONCRETE EXPOSURE	MEMBER	REINFORCEMENT	SPECIFIED COVER (mm)
SUBSTRUCTURE			
CONCRETE POURED AGAINST EARTH AND CONCRETE NOT FORMED IN CONTACT WITH EARTH	FOOTINGS	ALL	75
	RAFT FOUNDATION		
FORMED CONCRETE IN CONTACT WITH EARTH	INT. SIDES OF FOUNDATIONS	ALL	50
	GROUND BEAMS		
CONCRETE IN CONTACT WITH GROUND	SOG BASEMENT WALLS, AND ANY UNDERGROUND MEMBER PROTECTED BY WATERPROOFING AND BLINDING	> T16	50
		≤ T16	40
SUPERSTRUCTURE			
CONCRETE EXPOSED TO WEATHER	OPEN PODIUMS	> T16	50
		≤ T16	40
CONCRETE NOT EXPOSED TO WEATHER OR IN CONTACT WITH GROUND	SHEAR WALLS	ALL	40
	COLUMNS		
	BEAMS		
	NON PRE-STRESSED SLABS / JOISTS		
		≤ T32	25

3. T16 AND LARGER REINFORCING BARS SHALL NOT BE SPLICED EXCEPT AS LOCATED AND DETAILED ON THE DRAWINGS. T12 AND SMALLER BARS WITH LENGTHS NOT SHOWN SHALL BE CONTINUOUS. PROVIDE CLASS 'B' SPLICE UNLESS NOTED OTHERWISE. VERTICAL BARS SHALL NOT BE SPLICED EXCEPT AT HORIZONTAL SUPPORTS, SUCH AS FLOOR OR ROOF, UNLESS DETAILED OTHERWISE. ALL BARS ENDING AT THE FACE OF A WALL, COLUMN, OR BEAM SHALL EXTEND TO WITHIN 50 mm OF THE FAR FACE AND HAVE A 90 DEGREE HOOK, UNLESS OTHERWISE SHOWN.

4. BARS SHALL BE FIRMLY SUPPORTED AND ACCURATELY PLACED AS REQUIRED BY THE ACI STANDARDS, USING TIE AND SUPPORT BARS IN ADDITION TO REINFORCEMENT SHOWN WHERE NECESSARY FOR FIRM AND ACCURATE PLACING. PROVIDE DOWELS TO MATCH ALL REINFORCEMENT AT FOUR JOINTS, UNLESS SHOWN OR NOTED OTHERWISE. ALL DOWELS AND BOLTS SHALL BE ACCURATELY SET IN PLACE BEFORE PLACING CONCRETE. NO WELDING OF REINFORCEMENT (INCLUDING TACK WELDING) SHALL BE DONE UNLESS SHOWN ON THE DRAWINGS OR APPROVED BY THE ENGINEER. ALL SLAB AND BEAM REINFORCEMENT SHALL BE CHAIRED UP.
5. ALL BARS INTERRUPTED BY STRUCTURAL STEEL SHALL EXTEND TO WITHIN 25 mm OF STRUCTURAL STEEL FLANGE OR WEB AND HAVE A 90 DEGREE HOOK, UNLESS OTHERWISE SHOWN.
6. DRAWINGS SHOW TYPICAL REINFORCING CONDITIONS. CONTRACTOR SHALL PREPARE DETAILED PLACEMENT DRAWINGS OF ALL CONDITIONS SHOWING QUANTITY, SPACING, SIZES, CLEARANCES, LAPS, INTERSECTIONS, AND COVERAGE REQUIRED BY THE STRUCTURAL DETAILS, APPLICABLE CODE, AND TRADE STANDARDS. CONTRACTOR SHALL NOTIFY REINFORCING INSPECTOR OF ANY ADJUSTMENTS FROM TYPICAL CONDITIONS WHICH ARE PROPOSED IN PLACEMENT DRAWINGS TO FACILITATE FIELD PLACEMENT OF REINFORCING STEEL AND CONCRETE.
7. ALL PRINCIPAL REBAR SHALL TERMINATE WITH A STANDARD HOOK MINIMUM UNLESS SPECIFICALLY DETAILED OTHERWISE. REBAR BENDS SHALL BE MADE COLD. REBAR SHALL NOT BE BENT AFTER ANY PORTION OF THE BAR IS ENCASED IN CONCRETE.
8. ALL LAP SPLICES ARE CLASS 'B' LAP SPLICES UNLESS NOTED OTHERWISE.
9. MECHANICAL COUPLERS SHALL BE TYPE 1 ACCORDING TO THE REQUIREMENT OF ACI 25.5.7. AND SHOULD DEVELOPED A MINIMUM STRENGTH OF 1.25F_y IN CASE OF COUPLERS USED IN A TENSION MEMBER, ADJACENT BARS COUPLER SHOULD BE STAGGERED WITH A MINIMUM OF 75 MM.
10. ALL WALL FOOTING REINFORCEMENT SHALL BEND AROUND ALL CORNERS AND EXTEND 36 BAR DIAMETERS OR 450 MM WHICHEVER IS LARGER. UNLESS NOTED OTHERWISE.
11. ALL SLABS ON GRADE LESS THAN 150 mm IN THICKNESS SHALL BE REINFORCED WITH T12 REBARS AT 350 mm ON CENTERS EACH WAY, UNLESS NOTED OTHERWISE. PROVIDE ONE (1) LAYER OF 6x6W2.9XW2.9 WELDED WIRE FABRIC CONTINUOUS FOR EVERY 75 mm ARCHITECTURAL CONCRETE FILLS ABOVE THE STRUCTURAL SLAB.
12. ALL MECHANICAL PLUMBING AND ELECTRICAL EQUIPMENT PADS LESS THAN 100 mm THICK SHALL BE REINFORCED WITH AT LEAST ONE (1) LAYER OF 6x6W2.9XW2.9 WELDED WIRE FABRIC AND HAVE HOOKED DOWELS (T10 AT 300 mm ON CENTERS) INTO THE STRUCTURAL SLAB, UNLESS NOTED OTHERWISE. FOR PADS GREATER THAN 100 MM THICK, USE REINFORCING AS SHOWN IN THE TYPICAL DETAILS.
13. ADDITIONAL REINFORCEMENT SHALL BE PROVIDED AROUND ALL SLAB AND WALL OPENINGS INCLUDING DIAGONAL BARS WITHOUT EXCEPTION. IN CASE THIS NOTE CONFLICTS WITH TYPICAL DETAILS, CONTRACTOR SHALL CONFIRM WITH THE ENGINEER BEFORE EXECUTION.
14. ALL STRUCTURAL CONCRETE ELEMENTS REQUIRE REINFORCEMENT SINCE NO PLAIN CONCRETE ELEMENTS ARE USED. ALL CONCRETE SLABS SHALL HAVE A MINIMUM REINFORCEMENT OF 0.18% A_c EACH WAY CONTINUOUS.
15. REINFORCING STEEL SHOP DRAWINGS SHALL INCLUDE SLAB OPENINGS, DEPRESSIONS, SLOPES, CURBS, DRAINS, AND SLAB EDGE LOCATIONS FROM ALL MEP TRADES. GENERAL CONTRACTOR IS RESPONSIBLE TO COORDINATE AND INDICATE ALL SLAB OPENINGS ON PLAN AND SUBMIT TO THE STRUCTURAL ENGINEER PRIOR TO SUBMITTAL OF REINFORCING SHOP DRAWINGS.
16. INSTALLED REBAR IN CONCRETE SHALL NOT BE DAMAGED OR CUT BY CORING, DRILLING, SAW CUTTING, ETC
17. FOR REBARS' DEVELOPMENT LENGTH, LAP SPLICE AND HOOKS DETAIL 9 / SHEET S-010 SHOULD BE FOLLOWED. ALTERNATIVELY CONTRACTOR COULD SUBMIT MORE DETAIL CALCULATION ACCORDING CHAPTER 25 OF THE LATEST VERSION ACI 318.
18. REINFORCING STEEL BARS CALLOUTS SHALL BE READ AS SHOWN BELOW



E. FOUNDATIONS & EXCAVATION

1. THE DESIGN OF THE FOUNDATION SYSTEM IS BASED UPON THE CRITERIA AND RECOMMENDATIONS CONTAINED IN THE SOIL INVESTIGATION REPORT TITLED PDS/27401 REV.01 PREPARED BY PIONEERS LABORATORY FOR SOIL AND MATERIAL TESTING L.L.C. DATED 27/02/2025.
2. THE SOIL INVESTIGATION REPORT IS PART OF THE CONSTRUCTION SET, AND ITS RECOMMENDATIONS SHALL BE FOLLOWED AND SHALL BE CONSIDERED MINIMUM REQUIREMENTS UNLESS MORE STRINGENT REQUIREMENTS ARE PRESENTED IN THE SPECIFICATIONS OR ON THE DRAWINGS.
3. THE CONTRACTOR SHALL TAKE FULL ACCOUNT OF THE SOIL INVESTIGATION INFORMATION AND HIS OWN EXPERIENCE TO DESIGN THE NECESSARY TEMPORARY WORKS. THE DESIGN TOGETHER WITH THE SHOP DRAWINGS SHALL BE SUBMITTED FOR THE ENGINEER'S APPROVAL PRIOR TO DEWATERING. ALL BACKFILLING SHALL BE COMPACTED TO AT MAXIMUM DENSITY AND IN ACCORDANCE WITH THE SPECIFICATIONS.
4. EXISTING WATER TABLE IS AS PER SOIL REPORT
5. THE ALLOWABLE SOIL BEARING PRESSURES PER SOILS REPORT ARE AS FOLLOWS:

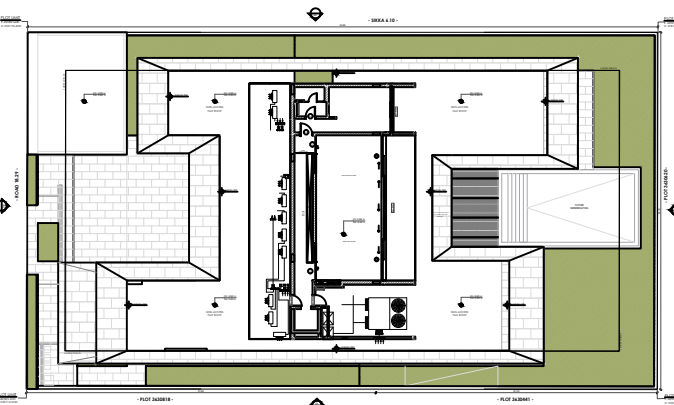
A. ISOLATED/COMBINED FOOTINGS 180 MPa

B. RAFT FOUNDATION 180 MPa
6. REMOVE LOOSE SOIL AND STANDING WATER FROM FOUNDATIO EXCAVATIONS PRIOR TO PLACING CONCRETE. THE GEOTECHNICAL ENGINEER SHALL INSPECT AND APPROVE ALL EXCAVATIONS. SOIL COMPACTION WORK PRIOR TO PLACEMENT OF ANY REBAR OR CONCRETE, SHORING INSTALLATIONS, BACKFILL MATERIALS AND BACKFILLING PROCEDURES.
6. LOCATE AND PROTECT EXISTING UTILITIES TO REMAIN DURING AND/OR AFTER CONSTRUCTION.
7. REMOVE ABANDONED FOOTINGS, UTILITIES, ETC. WHICH INTERFERE WITH NEW CONSTRUCTION, UNLESS OTHERWISE INDICATED.
8. NOTIFY THE ENGINEER IF ANY BURIED STRUCTURES NOT INDICATED, SUCH AS CESSPOOLS, CISTERNS, FOUNDATIONS, ETC. ARE FOUND.
9. THE CONTRACTOR IS SOLELY RESPONSIBLE FOR EXCAVATION PROCEDURES INCLUDING LAGGING, SHORING, CASING, UNDERPINNING, AND PROTECTION OF EXISTING CONSTRUCTION.
10. ALL EXCAVATIONS SHALL BE CARRIED OUT IN DRY CONDITIONS. THE CONTRACTOR SHALL BE RESPONSIBLE OF DEWATERING WHEN SO REQUIRED PER SOILS REPORT OR AS WARRANTED BY FIELD CONDITIONS.
11. PLACE BACKFILL BEHIND RETAINING WALLS AFTER CONCRETE HAS ATTAINED FULL DESIGN STRENGTH. BRACE BUILDING AND PIT WALLS BELOW GRADE FROM LATERAL LOADS UNTIL ATTACHED FLOORS AND SLABS ON GRADE ARE COMPLETE AND HAVE ATTAINED FULL DESIGN STRENGTH.
12. WHEN EXCAVATING TO FOUNDATION LEVEL CARE SHOULD BE TAKEN NOT TO DISTURB THE UNDERLYING MATERIAL. ALL FOUNDATION EXCAVATIONS SHALL BE INSPECTED AND ANY SPOTS REMOVED AND REPLACED WITH BLINDING CONCRETE, WHICH SHALL BE BEFORE CONSTRUCTION PROCEEDS. EXCAVATION SHALL AT ALL TIMES BE CARRIED OUT IN DRY CONDITIONS. ANY LOOSE OR WEATHERED ROCK IS TO BE REMOVED AND REPLACED WITH A MIX OF BLINDING CONCRETE TO THE REQUIRED FORMATION LEVEL.
13. DEWATERING IS EXPECTED TO BE REQUIRED UNTIL THE COMPLETION OF GROUND FLOOR SLAB. THIS SHALL BE CONFIRMED IN THE FOLLOWING DESIGN STAGE. DEWATERING SHALL NOT BE DISCONTINUED WITHOUT WRITTEN APPROVAL OF THE ENGINEER.
14. ANY OVER-CUT IS TO BE BACK-FILLED TO THE REQUIRED FORMATION LEVEL WITH BLINDING CONCRETE. THE CONTRACTOR IS TO ENGAGE THE SERVICES OF A SUITABLY QUALIFIED GEOTECHNICAL ENGINEER TO INSPECT ALL CUTS/ EXCAVATIONS AND FORMATION LEVELS, ENSURE THESE ARE CLEAN AND FREE FROM DEBRIS, AND ARE SUITABLE AS BEARING STRATA PRIOR TO COMMENCING OF THE FOUNDATION WORKS.
15. THE GEOTECHNICAL ENGINEER IS TO ENSURE THAT ANY JOINTS OR FISSURES IN THE UNDERLYING ROCK DOES NOT CAUSE SLOPE FAILURE OR REDUCE THE ALLOWABLE BEARING CAPACITY OF THE FOUNDING STRATA. SHOULD THE PROPOSED CUTS IN THE SLOPES RESULT IN THE INSTABILITY OF THE SLOPE OR INADEQUACY OF THE BEARING STRATA TO MEET DESIGN REQUIREMENTS, THE CONTRACTOR IS TO INFORM THE ENGINEER IMMEDIATELY.
16. ALL FOUNDATIONS, GROUND BEAMS, GROUND SLABS SHALL BE PLACED ON A MINIMUM THICKNESS OF 50mm OF GRADE C12/15 CONCRETE BLINDING UNLESS NOTED OTHERWISE.

GENERAL NOTES

1. This Drawing Issue has been checked and approved in accordance with the LOCI Quality Management System.
2. All works shall be carried out in accordance with particular B general specifications, in conjunction with relevant conditions of contract.
3. This Drawing shall not be used directly for Construction, but for preparation of Shop Drawings in coordination with other disciplines and Suppliers prior to manufacture, delivery and execution on site.
4. All dimensions and levels are in millimeter, unless noted otherwise, only written dimensions to be followed and not scaled from the drawing.
5. All dimensions to be verified at site and coordinated with other disciplines/ manufacturers in the preparation of shop drawings. Any changes shall be validated by the consultant.
6. For elements with indicative design, the contractor / supplier shall submit complete construction details and technical specifications for approval prior to execution, shop drawings to particularly incorporate details at: structural expansion joints as well as expansion / contraction joints needed for the respective materials/ finishes / related works.
7. Safety standards and statutory regulations shall be adhered to in the manufacture, supply, storage and installation of all works.
8. Any discrepancy or ambiguity whatsoever with dimensions / details / legends / specifications / etc, shall be brought to the attention of the Consultant and clarification obtained prior to finalization of tender / construction.

KEY PLAN



DRAWING REVISIONS

REV	DATE	DESCRIPTION	CHECKED	APPROVED
00	14.03.2025	ISSUED FOR APPROVAL	AN	BR

CLIENT

TARUN HARISH CHANDRA KHANNA

DRAWING DETAILS

GENERAL NOTES - (SHEET 1 OF 2)

SHEET #

269-S-001

REVISION

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ISSUE DATE:

14-03-2025

SCALE @ A1

AS SHOWN

PROJECT DETAILS

KHANNA VILLA



PROJECT LOCATION

PLOT NO. 3630160
Al Manara,
Dubai

PROJECT NUMBER

269

PROJECT PHASE

DETAIL DESIGN

CONSULTANT

LOCI
لوکار

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